

Final-Product Requirements Document (PRD v3.0)

One-sentence goal: Deliver a **single, final, authoritative PRD** for **LexCowork AI** that engineering, design, security, and investors can all execute against—no loose ends.

Below is the **FINAL PRD (v3.0)**, consolidating **all prior artifacts**, Cowork-parity verification, Super Admin governance, RAG, legal LLMs, guardrails, and the complete tech stack. This PRD supersedes earlier drafts and incorporates the verified Cowork-clone components from your uploaded documents .

Product Requirements Document (PRD v3.0)

Product Name: LexCowork AI

Tagline: *A Cowork-class legal productivity platform for India & the USA—governed, grounded, and safe by design.*

1. Product Vision

LexCowork AI is a **Cowork-style, action-oriented legal platform** that enables users to perform structured legal tasks—contract review, drafting, compliance, and research—using **agentic workers, RAG-grounded intelligence**, and **human-approved actions**, all under **strict legal guardrails**.

Unlike general AI copilots, LexCowork AI is:

- **Legal-only** (hard domain constraints)
- **Jurisdiction-aware** (India + USA)

- **Auditable and enterprise-ready**
 - **Non-advisory by default**
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2. Target Users

User	Primary Use
Founders / SMBs	NDA's, MSA's, policies, compliance
In-house legal teams	First-pass review, drafting
Legal consultants	Research memos, intake
Attorneys (assistive)	Redlines, clause comparison

 Not a replacement for licensed legal counsel.

3. Roles & Access Model

Roles

- **Super Admin (single / limited):** Platform owner & risk governor
- **User:** Executes legal tasks via workers

Super Admin Capabilities

- Enable/disable workers
- Control jurisdictions (US/IN + states)
- Upload, approve, revoke RAG sources
- Approve clause libraries/templates
- Control action permissions
- View all audit logs
- Kill-switch: pause tasks, freeze writes

Super Admin **cannot** bypass legal guardrails or enable legal advice.

4. In-Scope vs Out-of-Scope

In Scope

- Contract review & redlining
- Clause explanation & comparison
- Policy drafting (Privacy, HR, Terms)
- Compliance checklists (DPDP, CCPA, GDPR, SOC2)
- Legal research summaries
- Intake & issue spotting (non-advice)

Out of Scope (Hard Block)

- “Should I sue/file/register?”
 - Court strategy or outcome prediction
 - Guaranteed legal outcomes
 - Criminal or regulatory evasion
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5. Jurisdiction Support (Core Differentiator)

Phase-1 Regions

-  USA: Federal + TX, CA, NY
-  India: Central laws (Companies Act, IT Act, DPDP Act)

Jurisdiction Router (Mandatory)

Every task must resolve:

```
{  
  "country": "US | IN",  
  "state": "TX | CA | NY | null",  
  "confidence": "explicit | inferred"  
}
```

If ambiguous → ask **one** clarification question.

6. Cowork-Style Architecture (Cloned & Legalized)

6.1 Workers (Agentic)

- Orchestrator Worker (mandatory)
- Contract Review Worker
- Policy Drafting Worker
- Compliance Worker
- Research Memo Worker
- Intake Worker

Each worker includes:

- System rules
 - Sub-agents
 - RAG hooks
 - Verification pass
 - Output formatter
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6.2 Orchestrator Worker

Responsibilities:

- Resolve jurisdiction
 - Select workers
 - Run multi-step pipelines
 - Merge outputs
 - Enforce safety
 - Stop at human review for writes
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7. Task Lifecycle (Sync + Async)

States:

DRAFT → QUEUED → RUNNING → REVIEW_REQUIRED → COMPLETED → ARCHIVED

Execution Modes

- **Synchronous:** small reviews, clause explanation
- **Asynchronous:** large contracts, multi-doc RAG, audits

Rule: Async tasks may **never** auto-apply actions.

8. Legal Recipes (Reusable Workflows)

Pre-configured workflows containing:

- Jurisdiction
- Workers
- RAG scope
- Output formats
- Allowed actions

Only Super Admin can create/edit recipes.

9. RAG Architecture (Legal-Grade)

Objectives

- Eliminate hallucinations
- Ensure jurisdiction correctness
- Support enterprise knowledge

Sources

Primary (Admin-managed):

- Templates, clauses, policies, playbooks

Reference (read-only):

- Indian statutes & case law
- US federal/state references

Segmentation (Hard Rule)

/rag/us/federal

/rag/us/state/tx

/rag/in/central

No cross-jurisdiction retrieval unless explicitly requested.

Retrieval Flow

1. Jurisdiction resolved

2. Scoped retrieval
 3. Verification pass
 4. Citations attached
 5. No source → no claim
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10. Legal LLM & Intelligence Stack

Core Reasoning

- **Anthropic Claude API / Workers** (primary)

Legal-Trained / Legal-Database APIs (Pluggable)

- **Indian Kanoon API / kanoon.dev** (India case law)
- **vLex Developer APIs** (multi-jurisdiction)
- **LexisNexis APIs** (enterprise, optional)
- **Thomson Reuters APIs / CoCounsel** (enterprise, optional)

Claude remains the **orchestrator**; legal APIs act as **grounding sources**, not decision engines.

11. Safe Actions (Cowork-Style, Controlled)

Allowed Actions

- create_document
- insert_redlines
- generate_summary
- generate_checklist
- compare_versions

Rules

- Preview required
- Diff view mandatory
- Explicit user approval
- Fully logged

- ✗ No silent edits
- ✗ No deletes without admin override

12. Legal Workspace

- Versioned documents
 - Jurisdiction-tagged files
 - Diff history
 - Export (PDF/DOCX)
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13. Guardrails & Compliance

3-Layer Gates

1. **Policy Gate:** blocks advice, court strategy
2. **Tool Gate:** allowlist + approval for writes
3. **Output Gate:** jurisdiction tag + disclaimer + citations

Mandatory Footer

- Jurisdiction
 - “Informational only – not legal advice”
 - Confidence score
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14. Audit & Explainability

Logged for every task:

- User
- Workers used
- RAG sources
- Draft → final diff
- Approval events
- Timestamp & jurisdiction

Immutable, append-only logs.

15. Tech Stack (Final)

Frontend

- Next.js + TypeScript
- shadcn/ui + Tailwind
- TipTap editor + diff viewer
- Supabase Realtime

Backend

- Supabase Auth + RLS
- Supabase Postgres + pgvector
- Supabase Edge Functions
- Trigger.dev / Inngest (async)

Security

- RLS everywhere
 - Rate limiting (Upstash)
 - WAF (Cloudflare)
 - Secure headers, secrets server-only
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16. Non-Functional Requirements

Area	Requirement
Security	SOC2 / GDPR ready
Performance	<5s standard tasks
Isolation	Tenant-level RAG
Extensibility	New countries via packs
UI	Cowork-style task canvas

17. MVP Definition (90–120 Days)

Must-Have

- Super Admin console
- Orchestrator + 3 workers
- RAG ingestion & retrieval
- Legal workspace
- Audit logs

Phase-2

- Enterprise legal APIs
 - Attorney handoff
 - Jurisdiction comparison
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18. Strategic Positioning

LexCowork AI is **not**:

- A legal advice system
- A court strategy engine

LexCowork AI **is**:

- A Cowork-class legal productivity platform
 - Governed, grounded, enterprise-safe
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FINAL STATUS

This PRD is **implementation-ready**, **investor-ready**, and **security-defensible**.

If you want, next I can:

- ① Convert this into a **system architecture diagram**
- ② Produce a **Sprint-ready engineering backlog**
- ③ Create an **investor pitch deck**

Just say the word.